

CSE - 641 Computer Vision

Weekly Project Report – 6



**Ahmedabad
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Genotype-to-Phenotype Modeling of Alzheimer's Disease Using MRI Radiomics and Polygenic Risk Scores

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Abstract—Alzheimer disease is a disorder that affects the brain leading to loss of memory and progressive impairment of thinking ability. Studies indicate that genetic factors as well as changes in brain structure are significant in the pathogenesis of Alzheimer disease. However, the correlation between inheritable genetic risk and structural brain change is not fully understood.

This project examines the association between genetic vulnerability and brain structure using Polygenic Risk Scores (PRS) and MRI-based radiomic features. Important brain regions including the hippocampus and entorhinal cortex are analyzed using MRI scans from the Alzheimer Disease Neuroimaging Initiative (ADNI) dataset.

PRS values are computed to estimate inherited genetic risk, and machine learning models are used to identify relationships between genetic risk and MRI features. The goal is to better understand the role of genetic factors in early brain changes related to Alzheimer disease, potentially before clinical symptoms appear.

I. INTRODUCTION

Alzheimer disease is an advanced disorder of the brain which gradually impairs the memory, ability to think and general brain performance. It is connected with when brain tissue is gradually lost throughout time. These changes are usually studied by structural magnetic resonance imaging (MRI), particularly in those parts of the brain that become affected at an early stage in the disease, the hippocampus, entorhinal cortex and temporal lobes [1]. Meanwhile, genome wide association studies of large sizes have enhanced the insights into the genetic etiology of Alzheimer. The genetic risk variants that have been identified through such studies are too many and predispose an individual with the probable cause of the disease [2]. These variants may be used together in a polygenic risk score (PRS) i.e., to sum up inherited genetic risk in a single value [3]. Both MRI markers and PRS can be used to understand the disease, but this is in two dimensions- brain structure and genetic risk. There have been many research studies that attempted to improve early detection of Alzheimer disease with use of brain imaging methods. Structural magnetic resonance imaging is one of the methods that have been used majorly as it is capable of showing the changes that take place in the brain regions which are affected by disease. One of the affected regions is the hippocampus which is part of the brain that is affected in the Alzheimer disease as it is the part associated with memory. Previous studies were interested in the size of the brain regions. However with the use of the radiomics method, more quantitative features of the MRI images may be obtained that may potentially reveal more details. The authors of the paper [5] would like to create a model of radiomics using MRI images of the hippocampus of the brain to aid in the identification of patients suffering from Alzheimer disease. This would be achieved by the use of data from various cohorts to ensure the reliability of the model. Radiomic features would be achieved by the use of MRI images of the hippocampus. This would then be used to create a model that would aid in the differentiation of patients suffering from Alzheimer disease and healthy patients. This would be achieved by the use of MRI images and the expression of genes present in the blood.

This would then attempt to establish a connection between the genetic features and the biological processes aiming to have a better understanding of the changes taking place in the brain as shown by MRI images.

Recent studies have explored radiomics-based analysis of MRI scans to extract quantitative features that may reveal subtle changes not visible through conventional measurements. For example, Xia et al. [5] developed a hippocampal radiomics model using multi-cohort MRI data to differentiate Alzheimer patients from healthy controls and to investigate biological mechanisms linked to imaging features.

Combining radiomic features with genetic information may help improve early detection and understanding of disease progression. The summary of the selected parameters from the ADNI database is given in the Table 1.

II. DATASET DESCRIPTION

The data used in this project is obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) a multi-centered longitudinal large scale study established to identify and validate biomarkers for the early detection and monitoring of Alzheimer's disease progression. This data was acquired following strict protocol guidelines to ensure consistency and reproducibility.

Data in this study has been taken from both ADNI2 and ADNI3 phases. Both phases involve longitudinal data that involves several repeat examinations per patient. Therefore in contrast to a cross-sectional design where only a single instance is considered, a longitudinal design is used here which involves multiple instances per subject.

The study participants are categorized into these primary groups:

- Cognitively Normal (CN)
- Mild Cognitive Impairment (MCI)
- Alzheimer's Disease (AD)

Data available through ADNI include:

- Structural MRI
- PET imaging (FDG-PET and Amyloid PET)
- Cerebrospinal fluid biomarker levels
- Genomic data (APOE genotype)
- Neuropsychological test results (MMSE, CDR, ADAS-Cog)

The filtering of the dataset depends on ADNI2 and ADNI3 phases while the visits used are various these include baseline and follow up visits that are spread throughout different periods of time to analyze disease progression.

The following criteria are taken into account in the selection of the images in this study, Image Description-MPRAGE, Weighting-T1 weighted MRI, Acquisition-3D, Field Strength-3T.

The dataset is cross-standardized across acquisition sites, ensuring consistency and reliability. Due to its comprehensive and longitudinal nature the ADNI dataset is widely used in machine learning and deep learning research for classification, progression prediction and biomarker discovery in Alzheimer's disease studies.

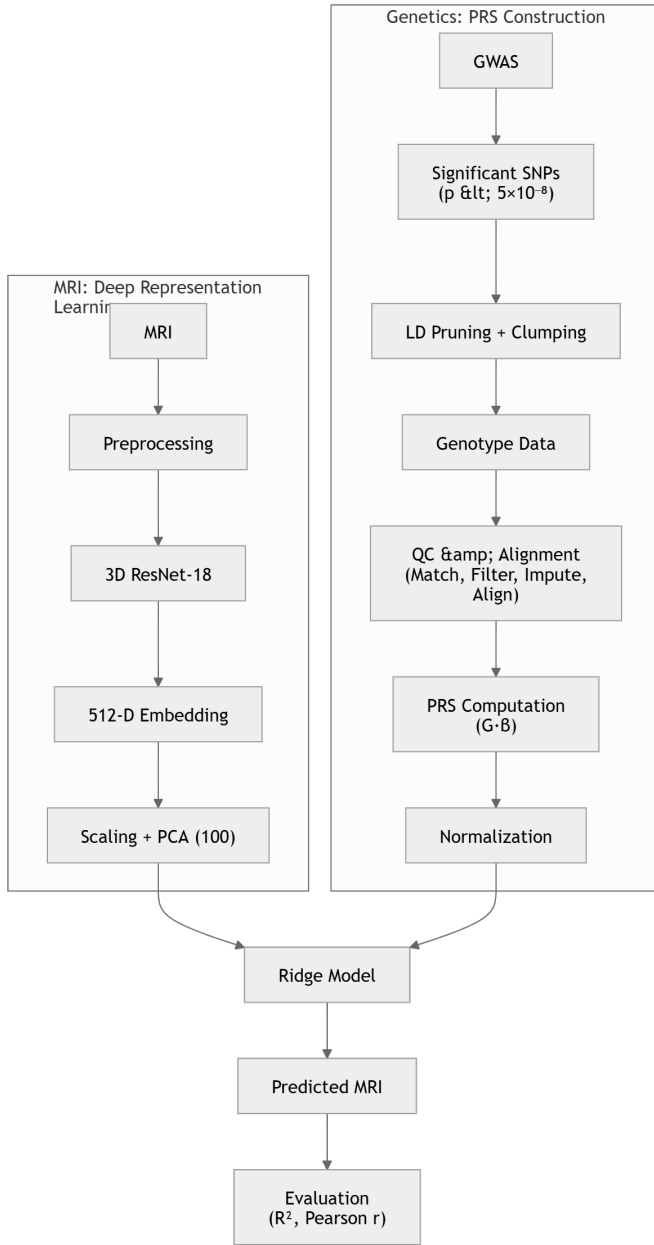


Fig. 1: Flowchart of the performed end-to-end pipeline.

III. LITERATURE REVIEW

A. Longitudinal structural MRI-based deep learning and radiomics features for predicting Alzheimer's disease progression

The authors in this paper show how a longitudinal MRI-based paradigm can be utilized to forecast the subjects who will progress to Alzheimer disease from mild cognitive impairment (MCI). They build a twostage deep learning model on the T1-weighted images of the Alzheimer Disease Neuroimaging Initiative dataset. They first train a 3D ResNet18 model on individual visit MRI volumes to enable it to learn spatial brain features related to disease risk with a survival-based ranking loss. The learned feature representations are then

used as input to a time-varying LSTM that trains to learn multiple MRI scans over time, accounting for irregular time intervals between visits. The attention mechanism identifies the most relevant time points. They also discard gray matter radiomics features like shape, intensity, and texture features, and discard them using PCA, and then combine them with deep features. The final model is being placed within a survival analysis framework to estimate individualized risk of time-to-conversion, rather than just binary classification.

B. Deep Learning-Based Feature Extraction with MRI Data in Neuroimaging Genetics for Alzheimer's Disease

Another interesting research direction in imaging genetics involves the use of deep learning in order to generate biologically informative representations for use in genetic association studies. With regard to such efforts, Chakraborty et al [4] have introduced an algorithm that incorporates CNNs with GWAS analysis for studying Alzheimer's disease (AD). The major difference between this approach and conventional GWAS methodologies based on ROI-derived features is that in their work, CNNs are used to extract data-driven features from whole 3D MRI volumes; hence, no spatial information is lost. Furthermore, as the name suggests, such features reflect complex patterns of brain atrophy and can be considered as endophenotypes – intermediate quantitative traits that are more biologically representative than actual disease outcomes. Finally, the learned features can be used in GWAS analysis by relating the associated SNPs to each individual; the resulting methodology allows detecting genetic variations that impact brain structure, recovering previously known AD-associated SNPs as well as identifying novel associations. The combination of deep learning algorithms and statistical genetics is clearly advantageous, although this particular framework does not consider multimodal interactions explicitly and does not involve predictive modeling.

IV. METHODOLOGY

The goal of this project is to examine the relationship between genetic risks and brain structure by integrating Polygenic Risk Scores with features derived from MRI scans. First, datasets were collected that contained both genetic information and corresponding MRI scans. The individuals that were common to both datasets were identified using a common identifier, referred to as IID. These individuals were then used for analysis.

The next steps were to preprocess the genetic information to obtain PRS values, which represent the overall risk based on various genetic weights. The MRI scans were then preprocessed by extracting features, whereby each scan is converted into a fixed-length numerical vector that represents brain features.

Finally, a new dataset is formed by combining the preprocessed PRS values with MRI features based on subject ID. The overall goal is to create a predictive model that predicts MRI features using PRS values as input. The overall strategy is to use 5-fold cross-validation.

TABLE I: Summary of Selected ADNI Dataset Parameters

Parameter	Selection
ADNI Phases	ADNI-2 and ADNI-3
Imaging Modality	Structural MRI
MRI Sequence	T1-weighted MPRAGE
Field Strength	3 Tesla (3T)
Visit Type	Temporal
Subject Groups	CN, MCI, AD
Image Format	NIfTI (.nii)
Genetic Data	APOE Genotype

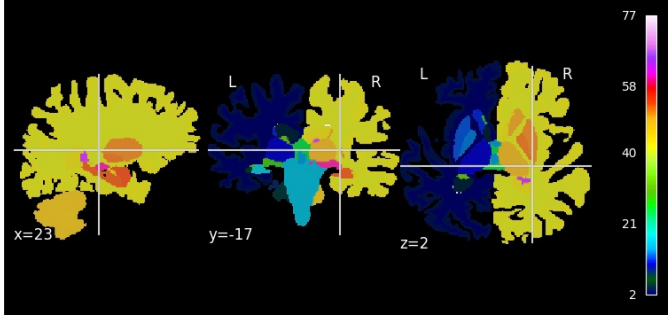


Fig. 2: Segmented 3D brain MRI highlighting different anatomical regions across multiple views

For each fold, MRI features are standardized using PCA to reduce dimensionality. A Ridge Regression model is then used to create a predictive model using PRS values as input features and MRI features as output features.

V. WORK COMPLETED IN WEEK-6

In this week's session, we merged datasets from Phase 2 and Phase 3 of ADNI, considering only those who have MRI scans as well as SNP data for conducting imaging-genetics analysis. The MRI scans, which are in NIfTI file format, were arranged and preprocessed for further processing tasks. Instead of using region-wise segmentation, we use data-driven methodology in which whole-brain representations are utilized.

Moreover, we explored different advanced methodologies, such as deep learning for MRI feature extraction, multimodal fusion techniques, and GWAS analysis based imaging-genetics framework. This helped us in designing our methodology to connect genomic and MRI feature data. The current phase mainly focused on preparing datasets, understanding various methods and pipeline development.

VI. PLAN FOR WEEK-7

For week 7, we will focus on adjusting our current pipeline on selected method and getting ready to apply temporal modeling:

- Improving the quality of the data set through addition of the longitudinal series of MRIs and aligning of patients due to multiple visits.
- We will do some modifications so that imaging and genomics data are integrated in order to fit the new modeling approach.

- The switching of the modeling that is performed with the help of only regional characteristics to the model that works with representation learning with CNNs.
- The temporal modeling methods will be applied to identify changes between multiple visits of patients.
- Validation of the new pipeline to make sure there is easy data flow for the model training/testing process.

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